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Exciton photoluminescence from ZnO layers produced by laser-induced gas breakdown processing

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ABSTRACT The plasma of optically-excited gas breakdown has been used to treat a Zn target in atmospheric pressure gases (air, O_2, N_2, Ar). The breakdown is produced near the target by a pulsed CO_2 laser radiation, yielding to a local erosion of the target under the irradiation spot and the formation of a porous nanostructured layer, consisting of ZnO nanoscale spheres. We show that the produced nanostructured layers exhibit an intense exciton emission band in the ultraviolet range (380–385 nm), while defect-related photoluminescent bands were weak and could be completely removed by varying the fabrication parameters. Properties of the produced layers were found to be very promising for the development of optoelectronic devices.

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1 Introduction

Zinc oxide is considered an extremely prospective semiconductor material for applications in optoelectronics [1]. The interest in this material is due to its direct wide band gap (3.37 eV at 300 K), the availability of fairly high quality single crystals and a large exciton binding energy of 60 meV. Along with its optical transparency, all these properties enable consideration of ZnO as an excellent candidate for the development of efficient UV emitters, flat panel displays etc. [1]. Furthermore, several recent studies have reported the possibility of creating a highly scattering and amplifying ZnO-based medium, leading to a mirror-less random lasing effect [2–6]. The applications just mentioned, require the fabrication of films with strong excitonic emission [1], which is not easy to achieve by current methods, including anodical etching [7], sol–gel procedures [8], spray pyrolysis [9, 10], rf magnetron sputtering [11, 12], metal-organic chemical vapor deposition [13], thermal evaporation [14], molecular beam epitaxy [15], laser ablation [6, 16–18] etc.. This stimulates the research of new fast, low-cost methods for the production of ZnO structures with desired characteristics.

In this paper, we report a local patterning of ZnO layers with strong exciton photoluminescence (PL) emission by applying methods of laser-induced air breakdown processing [19–22]. This processing technique uses a phenomenon of gas breakdown, which takes place during the focusing of pulsed laser radiation in atmospheric air and is accompanied by the production of hot and strongly absorbing plasma [23]. In contrast to conventional laser ablation in high pressure gases by nanosecond [24, 25] or femtosecond [26] laser pulses, which lead to the removal of a significant amount of material, and the production of a crater at relatively high fluences, this breakdown processing results in the formation of a highly porous structure without any significant material removal, as demonstrated in experiments with Si [19–21].

2 Experimental

A pulsed TEA CO_2 laser beam (wavelength 10.6 μm, pulse energy 1 J, pulse length 1 μs FWHM, repetition rate 3 Hz) was focused by a Fresnel lens with a focal length of 5 cm, giving a radiation intensity of about 10^8 W/cm^2 at the focal plane. Processing fluences between 0.14 and 0.4 J/cm^2 were varied by placing the 1×1 cm^2 metallic Zn target (purity 99.995%) in different positions from $z = -20$ mm to $z = 20$ mm, where $z = 0$ corresponds to the focal plane. Experiments were carried out in atmospheric air (1 atm, 20 °C, 40% humidity) or in atmospheric pressure gases (O_2, N_2 or Ar). An experimental chamber was turbopumped down to a base pressure of lower than 5×10^{-5} Torr before filling with gases. Photoluminescence spectra were recorded on a double spectrometer (U100, Instruments SA) using 325 nm radiation from a He-Cd$^+$ laser with a power density of 30 W/cm^2 as the source, and a GaAs photomultiplier as the detector. High resolution PL spectra were also recorded on an Acton 500i spectrometer coupled with a rapid iCCD camera using a 100 fs pulsed laser excitation at 355 nm. Field emission scanning electron microscopy (SEM) (JEOL 6320 F), was used to examine the structural properties of the films. The crystalline structure of the films was obtained by X-ray diffraction (XRD) spectroscopy (X'pert XRD system, Philips Corp). In addition, X-ray photoemission spectroscopy (XPS) was used to determine the surface layer composition.

3 Results

In our experiments, focusing the radiation on a Zn target led to the formation of a laser-induced gas breakdown

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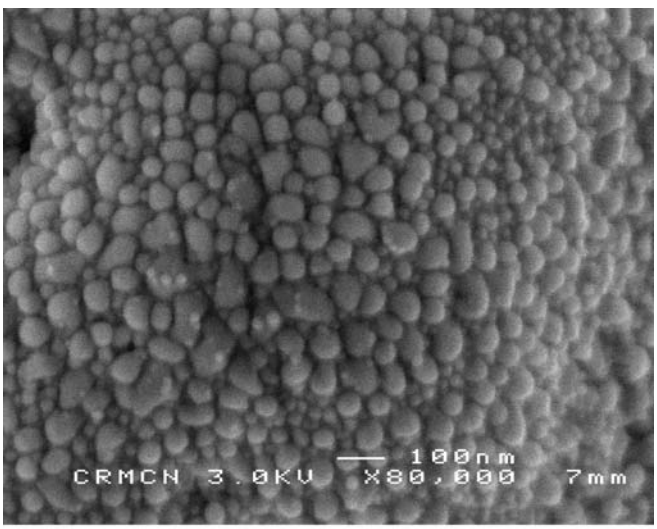
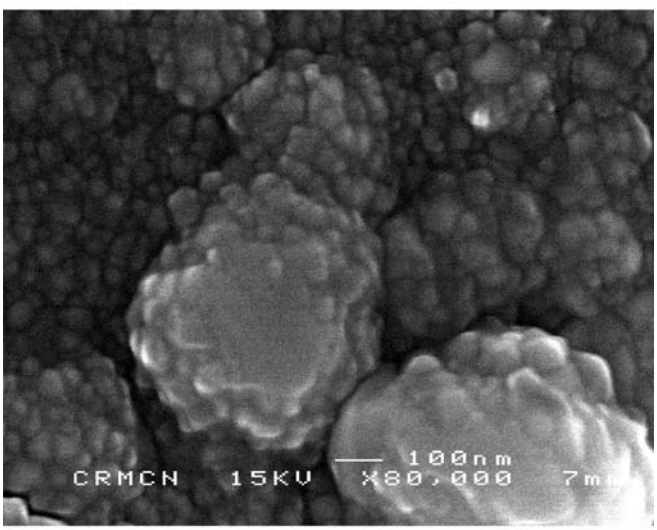
plasma, which manifested as a bright flash, moving from the target toward the focusing lens. After the breakdown ignition, the plasma intensity increased progressively with the number of laser shots and stabilized after only 5–10 shots. The intensity gain was apparently connected to the appearance of mechanical defects on the target surface, which improved the radiation absorption. All processing experiments were carried out near the threshold of plasma production to minimize the size of the treated area.

After the breakdown initiations by several laser pulses, a gray-tint area was formed under the focal spot on the target surface. The size of this area was in good correlation with the laser spot size, although certain surface modification around the spot could also take place under relatively high laser fluences. It is important to note that the crater area, usually surrounded by a hill of re-deposited material, was essentially absent, as this phenomenon takes place in experiments with conventional UV [24] or fs [26] laser ablation under the relatively high fluences of laser radiation.



a



b

FIGURE 1 Typical SEM images of a Zn target surface under the radiation spot after 10 breakdown initiations. Figures (**a**) and (**b**) correspond to laser fluences of 0.17 and 0.39 J/cm^2 during the treatment

Figure 1a shows a typical high resolution SEM image of the treated area under a moderate fluence of 0.17 J/cm^2. One can see that the layers represent a porous material, consisting of nanoscale particles with the size ranging from 20 to 50 nm. The nanoparticles are well packed, forming a multilayer photon crystal-like structure. As shown in Fig. 1b, the increase of the laser fluence to 0.39 J/cm^2 leads to the coagulation of nanoparticles and the formation of larger microscale spherical structures, containing the 20–50 nm nanoparticles. The formation of such structures is probably explained by a partial melting and re-crystallization of nanoparticle blocks under higher temperatures of laser plasma. Note that the structure presented in Fig. 1a is similar in some respects to layers of chemically synthesized micro- and nanostructured films, which were formed by pressing ZnO nanoparticle-based powders [3]. Such films were known to exhibit the random lasing effect.

Similar to the case of Si [22], the morphology of ZnO layers depended on the gas nature. The presence of a gas with relatively low ionization threshold maximized the intensity of the breakdown plasma, leading to a faster coagulation of nanoparticle blocks and the formation of structures shown in Fig. 1b.

XPS studies confirmed the formation of ZnO layer on the Zn substrate, which was illustrated by the presence of zinc $2p\,3/2$ and oxygen $1s$ peaks in XPS spectra. The treatment led to the shift of Zn $2p\,3/2$ peak from 1022 to 1021.4 eV and of the oxygen $1s$ peak from 531.8 to 529.9 eV, suggesting a transformation of surface layer from naturally surface-oxidized Zn to pure ZnO. Table 1 shows changes in the relative content of carbon, zinc and oxygen in the surface layer processed at different laser fluences. One can see that the treatment led to a drastic decrease of carbon content, associated with the removal of initial contaminants from the surface, and the increase of the Zn/O ratio. While not exceeding 0.24 for initial zinc samples, this ratio increased up to 1 when the laser fluence was relatively high (> 0.20 J/cm^2).

The processed area exhibited PL signals, which could be easily seen by the naked eye. PL spectra normally had a very strong band-edge emission band (380–385 nm), which is normally attributed to exciton transitions [1], and sometimes a defect-related band in the green-red range (500–650 nm). Typical PL spectra from samples produced under different laser fluences in air are shown in Fig. 2a. Here, one can see that to observe the generation of a defect-related band it is sufficient to treat the target near the plasma initiation threshold (0.11 J/cm^2), whereas the generation of the exciton-related band requires slightly higher fluences (> 0.14 J/cm^2). Max-

Treatment	C $1s$	Zn $2p\,3/2$	O $1s$
Zn before treatment	68	6	25
0.14 J/cm^2	52	14	34
0.16 J/cm^2	28	32	40
0.2 J/cm^2	34	32	34
0.25 J/cm^2	34	33	33
0.39 J/cm^2	32	34	34

TABLE 1 Relative concentrations of zinc, oxygen and carbon obtained from an XPS survey signal

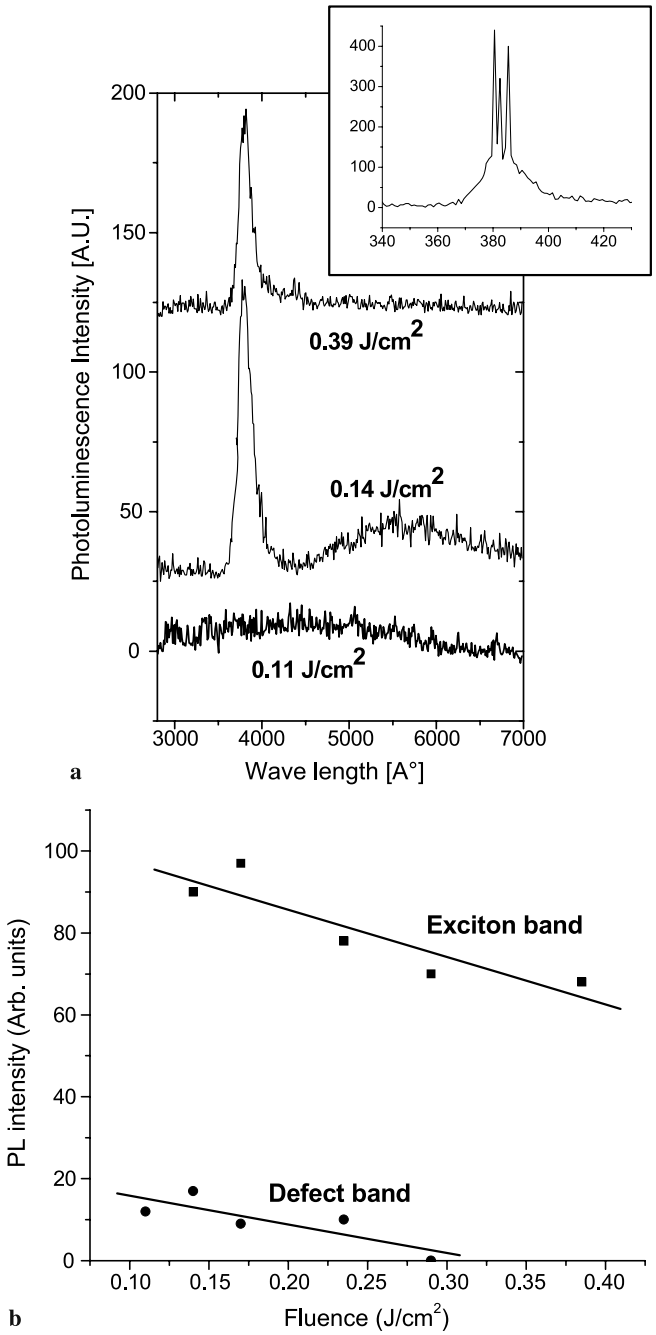


FIGURE 2 (**a**) Typical photoluminescence spectra from the ZnO layers produced by laser-induced breakdown processing of Zn in air at different fluences. The *inset* of figure (**a**) shows a high resolution spectrum illustrating the random lasing effect. (**b**) Dependence on the exciton and defect-related band on the fluence of laser used for breakdown initiation

imal values of PL for both bands were observed for treatment at 0.15–0.2 J/cm^2, while the absolute value of exciton-related PL was larger by about an order of magnitude. Further increase of the fluence led to a complete disappearance of defect-related bands and PL spectra were characterized by a single PL peak at 380–385 nm, as shown in Fig. 2a for the fluence of 0.39 J/cm^2. The exciton-related peak was very stable to laser illumination without any significant sign of degradation for many hours of continuous illumination. Such a strong and stable PL was a pleasant surprise, taking into account the extreme simplicity of the fabrication method, experimental conditions (ambient air) and starting material (a bulk piece of Zn). It should also be mentioned that the increase of the pumping laser intensity during PL measurements could lead to the appearance of narrow and intense lines within the exciton emission band, as shown in the insert of Fig. 2a. Notice that these lines are often attributed to the random lasing effect [2–6]. Details of the properties of the narrow lines are presented in a separate paper [28].

To understand conditions of formation of exciton and defect-related PL centers, we performed the treatment in the atmosphere of different gases (O_2, N_2, Ar, He). Typical PL spectra of layers produced in some of these gases are shown in Fig. 3. One can see that the exciton-based PL band appeared only when some amount of oxygen was present during the treatment process. In other words it means that these centers can not be formed during the post-fabrication natural oxidation of the layers produced in oxygen free atmosphere. The treatment in different gases could lead to the formation of defect-related PL centers. Their positions were different for these gases (425 nm for N_2; 510 for Ar and 560 for O_2),

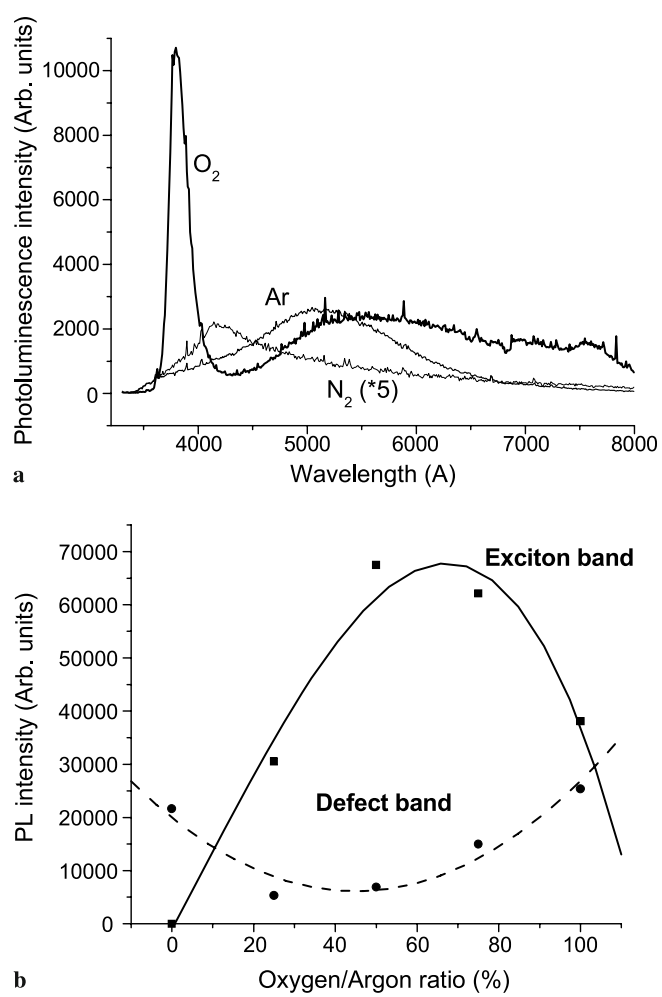


FIGURE 3 (**a**) Typical photoluminescence spectra from a Zn target after the laser induced breakdown processing of Zn in pure nitrogen, argon and oxygen. Data for nitrogen and argon were multiplied by a factor of 5. (**b**) Amplitude of the exciton and defect-related band as a function of the oxygen/argon ratio

suggesting that different breakdown plasma intensities could affect the formation of these centers. In addition, oxygen provided favorable conditions for the formation of these defect centers with PL amplitudes about 5-times higher then ones obtained on samples treated in N_2 and Ar. Surprisingly, air was more efficient than pure oxygen in the formation of layers exempt of defect-related PL band and this phenomenon was well repeatable. As shown in Fig. 2, this band could be completely removed by optimizing the laser fluence during the fabrication process, while such a removal was impossible in the case of pure oxygen (Fig. 3a). The PL intensity of the exciton band was also 3–10 times stronger for layers prepared in air, suggesting that a relatively high concentration of oxygen leads to a deviation from the optimal stoichiometry of ZnO-based compounds, which favors the defect band transition. As another test experiment, we used a mixture of O_2 and Ar, maintained at atmospheric air pressure. Figure 3b shows the intensity of the exciton- and defect-related PL bands as a function of the oxygen/argon ratio. Here, the intensity of the exciton-based PL increased with the increase of O_2 content upto a O_2/Ar ratio of about 0.6, while higher concentrations of O_2 led to a decrease of its intensity. The appearance of such an optimum was probably related to an optimal stoichiometry of Zn and O in the produced layers, providing the most efficient exciton transitions. One can also see from Fig. 3b that the optimum for defect-related bands was different reaching its minimum for a O_2/Ar ratio close to 0.25. Thus, the relative contribution of exciton- and defect-related bands can be controlled by changing the content of O_2 during the fabrication.

4 Discussion

It is known from the literature that the CO_2 laser–matter interaction is characterized by a strong photon absorption by the plasma itself [23]. The target serves to generate initial electrons and the formed plasma develops in the cold gas toward the focusing lens, and absorbs the main IR radiation power at its shock wave forefront through the inverse Bremsstrahlung mechanism [27]. As a result, the plasma gets heated up to temperatures of about 10^4 K, while the radiation power on the target surface decreases dramatically and the radiation-related ablation of material drops [27]. After the laser pulse, the plasma decays gradually in milliseconds, while containing intense currents (with magnitudes up to 10^6 A) [29] and various electromagnetic effects [30–32]. In our opinion, the action of the radiation leads to a localized melting and even flash evaporation of the target material. This process probably takes place only at some points, which causes the appearance of pores on the target surface. The laser-ablated material and the upper target layer are then heated by the hot CO_2 laser-induced breakdown plasma or its currents, in milliseconds, leading to both additional phase transformations and the initiation of chemical reactions in the plasma. Since the radiation is pulsed at 3 Hz, one can assume recrystallization or local vapor redeposition of the material during the off-times. The combined action of laser- and plasma-related processes gives rise to a modification of properties of Zn-based compounds formed on the surface and their nanostructuring. The mechanism of surface treatment in our experiments seems to be similar in many respects to that of the electric spark processing, which was also used for the fabrication of Si [33] and ZnO-based [34] layers. The layers fabricated by the electric spark processing in pure oxygen exhibited similar exciton and defect-related PL signals. However, the electric spark treatment procedure was much slower and took several hours to produce layers with efficient exciton-related PL.

Note that in contrast to laser-induced air breakdown processing, conventional laser ablation can not enable exciton PL in the absence of defect-related bands. Normally, as-prepared layers synthesized by this method exhibit a relatively weak exciton band and a relatively strong defect-related PL [17, 18]. The removal of the defect band and amplification of the exciton-based emission band, can be obtained either by the thermal post-fabrication annealing of films [18] or a significant heating of the substrate during the pulsed laser deposition process [4]. Such a difference in the properties of structures synthesized by these two methods might be explained by different conditions of nanostructure formation and treatment. The laser ablation leads to an efficient removal of target material, which then quickly cools down forming a thin nanostructured film. Such a modification of ZnO properties is probably insufficient to form defect-free structures. In contrast, the breakdown processing initiates hot plasma, which heats both ablated species and the target surface. Thus, the breakdown processing combines surface nanostructuring, material heating and relatively slow heating of the treated zone, giving rise to a desirable modification of ZnO-based structures.

5 Conclusion

In summary, laser-induced air optical breakdown has been produced on a Zn target for local modification of its material. We showed that the breakdown production led to the formation of a porous layer under the radiation spot, exhibiting strong and stable exciton-related PL. The PL signals associated with defects can be minimized and even completely removed by changing the laser fluence and varying the oxygen content. The layers produced are of importance for optoelectronic applications.

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